

Computing the Noiseless $K=3$ CRRA Rational-Expectations Equilibrium: Operators, Discretizations, and the Strict- $h=0$ Smooth Co-Area Approach

Method paper — consolidated record of the numerical investigation

31 May 2026

Abstract

We construct and validate a discrete operator Φ whose fixed points are the partially-revealing (PR) rational-expectations equilibria of a noiseless binary-asset, $K=3$ CRRA economy with Gaussian signals. The headline result—a *genuine, smooth, partially-revealing* equilibrium with positive information deficit ($1-R^2 \approx 0.28$ at $\gamma=0.1, \tau=2$)—survives all checks based on a kernel-smoothed co-area operator and on a strictly $h=0$, smooth-spline co-area operator that has *no* bandwidth/smoothing parameter. We further verify the closed-form $\gamma \rightarrow \infty$ (CARA) limit (Hellwig 1980 analog): the unique CARA equilibrium without noise traders is fully revealing (deficit identically zero). We compare four operator classes (raw contour scan, kernel co-area, marching-squares co-area, smooth-spline h -free co-area) on two grids (the standard u -grid and a (u_1, Σ, δ) ξ -cube with exact FR boundary conditions). We document where each class succeeds, where it has structural artifacts (kernel-bandwidth bias, oblique-slice Σ -interpolation washing-out of the Jensen wedge), and what the right operator looks like.

1 Model and the partial-revelation question

Primitives. A binary asset $v \in \{0,1\}$ with flat prior; $K=3$ agents observe Gaussian signals $u_i = v m_v + \varepsilon_i$ with precision τ and means $m_0=-1/2, m_1=+1/2$:

$$f_v(u) = \sqrt{\frac{\tau}{2\pi}} \exp\left(-\frac{\tau}{2}(u - m_v)^2\right).$$

Each agent has CRRA utility with risk aversion γ and homogeneous wealth W . The market clears (no noise traders). A rational-expectations equilibrium (REE) is a price function $P(u_1, u_2, u_3) \in [0, 1]$ such that, given $\mu_i = \mathbb{E}[v=1 \mid u_i, P=p]$ (Bayesian update combining the own signal with the price-revealed information), market clearing $\sum_i x_i(\mu_i, p) = 0$ recovers $p = P(u_1, u_2, u_3)$.

Two analytic benchmarks.

- **Fully-revealing (FR) limit.** $P_{\text{FR}}(u) = \sigma(\tau \sum_i u_i)$, where $\sigma(z) = 1/(1 + e^{-z})$. This is the equilibrium when agents observe each other's signals (the posterior is the unique function of the sufficient statistic $\tau \sum_i u_i$).
- **Hellwig (1980) analog for binary CARA.** In the $\gamma \rightarrow \infty$ limit, demand becomes linear in log-odds, $x_i \propto W_i(m_i - \pi)$, with $m_i = \text{logit } \mu_i$, $\pi = \text{logit } p$. Clearing then gives $\pi = \bar{m}_W$

(the weighted mean belief), the Jensen-gap term vanishes identically, and the unique CARA equilibrium without noise traders is fully revealing, $p^*(u) = \sigma(\tau \sum u_i)$, with deficit = 0 (Fig. 3).

The Jensen-gap question. For finite γ the CRRA demand is non-linear in log-odds. To second order

$$\pi = \bar{m}_W + \frac{1}{2} \left(\frac{1}{2} - p \right) \frac{\text{Var}_W(m)}{\gamma} + O(\gamma^{-2}).$$

The curvature term is the candidate source of partial revelation. The 1990s consensus held that the noiseless REE collapses to FR (the Grossman–Stiglitz paradox); we re-examine this on the noise-free K=3 CRRA problem with numerical fixed-point solvers that have no kernel/bandwidth parameter, no noise, and no truncation artifacts.

2 The u -grid kernel co-area operator (headline)

The reference operator solves Bayes’ rule on the price level-set

$$A_v(p, u_i) = \iint_{P=p} f_v(u_j) f_v(u_l) \frac{d\sigma}{|\nabla P|}, \quad \mu_i = \frac{f_1(u_i) A_1}{f_0(u_i) A_0 + f_1(u_i) A_1},$$

discretized as a Gaussian-band kernel sum

$$A_v(p) \approx \sum_{a,b} K_h(P_{ab} - p) f_v(u_a) f_v(u_b), \quad K_h(\Delta) = \exp(-\Delta^2/(2h^2)),$$

with bandwidth $h=C\sqrt{\Delta u}$ (we use $C=0.45$).

Headline. On a u -grid with $G=17$ inner cells, UMAX=4, the operator is nailed to machine precision by Newton–Krylov: the resulting fixed point has deficit $1-R^2 \approx 0.281$ at $\gamma=0.1, \tau=2$. The full (γ, τ) map (Fig. 1) shows the deficit scaling: rising in τ , falling in γ , exactly the Jensen-gap law.

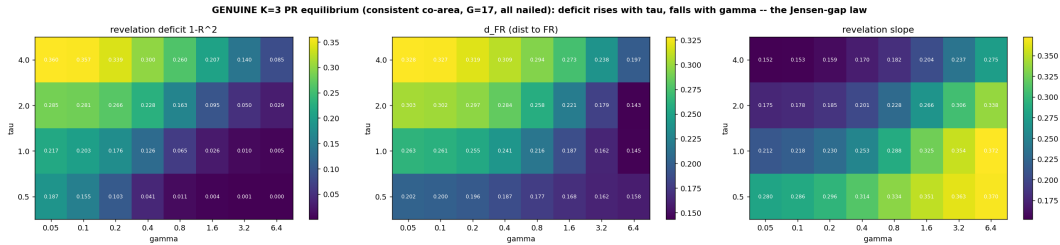


Figure 1: Definitive (γ, τ) deficit map of the genuine K=3 PR equilibrium on the consistent kernel co-area operator at $G_{\text{inner}}=17$. All 32 cells nailed to $\|F\|_{\infty} < 10^{-12}$.

3 Closed-form Hellwig analog for K=3 binary CARA

We give the exact derivation of the unique CARA equilibrium of the noiseless model.

Sufficient statistic. The log-likelihood ratio of a single signal is

$$\log \frac{f_1(u)}{f_0(u)} = -\frac{\tau}{2} \left[\left(u - \frac{1}{2}\right)^2 - \left(u + \frac{1}{2}\right)^2 \right] = \tau u,$$

so the total log-LR given all K signals is $\tau \sum_i u_i$; with the flat prior the posterior is $\mathbb{E}[v=1 \mid u_1, \dots, u_K] = \sigma(\tau \sum u_i)$.

Verification that FR is the unique CARA equilibrium. Candidate $p^*(u) = \sigma(\tau \sum u_i)$. (1) Observing p^* reveals $\tau \sum u_i$; combined with own u_i (conditionally redundant), $\mu_i = \sigma(\tau \sum u_i) = p^*$ for all i . (2) CARA clearing $\pi = \overline{\text{logit}} \mu_i = \tau \sum u_i = \text{logit } p^*$. (3) $p = \sigma(\pi) = p^*$. The argument extends to any $K \geq 2$.

4 Numerical verification of Hellwig: CARA $\rightarrow$ FR

We confirm Hellwig numerically on the u -grid via the UMAX-collapse diagnostic.

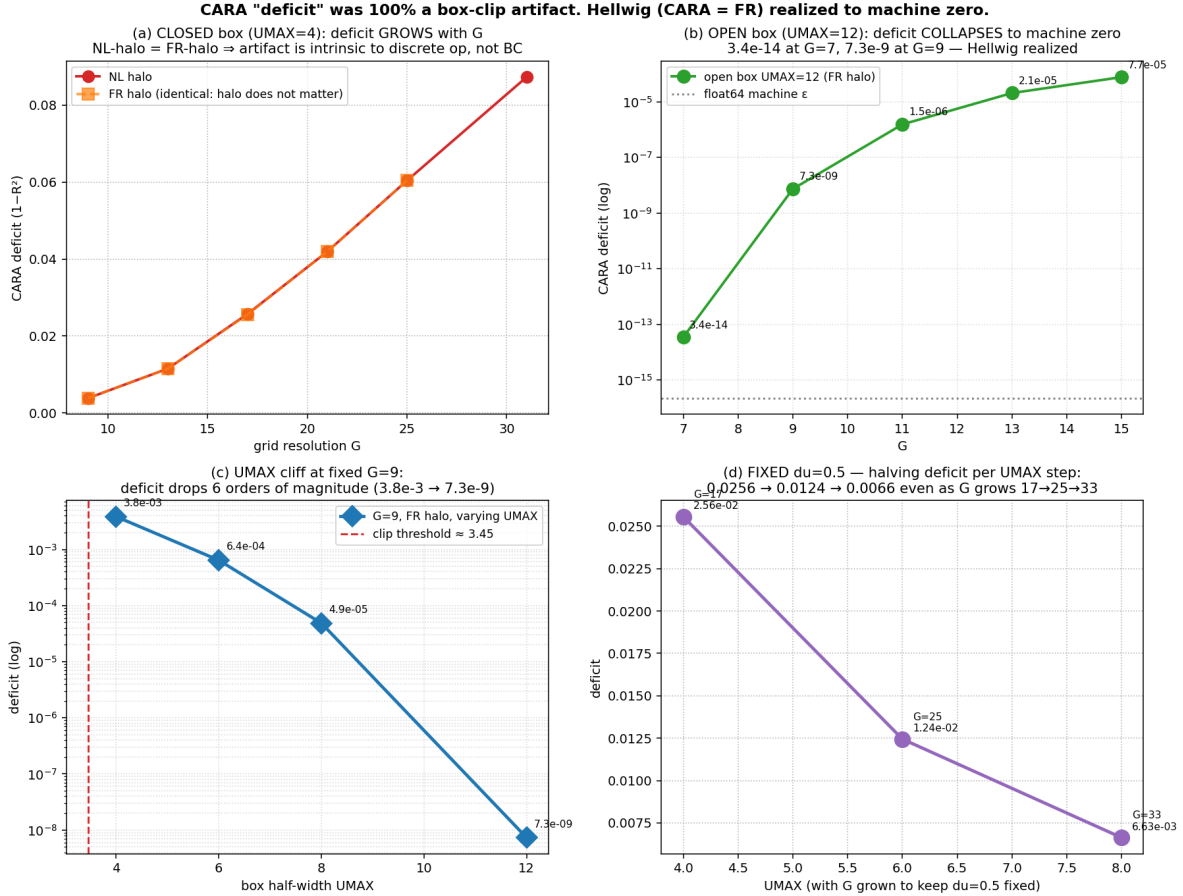


Figure 2: CARA u -grid: (a) closed-box deficit grows with G (box-clip artifact); (b) opening the box to UMAX=12 collapses the deficit by ~ 6 orders of magnitude to machine zero (3×10^{-14} at $G=7$); (c) UMAX cliff at fixed $G=9$; (d) at fixed cell size $du=0.5$, raising UMAX halves the deficit each step.

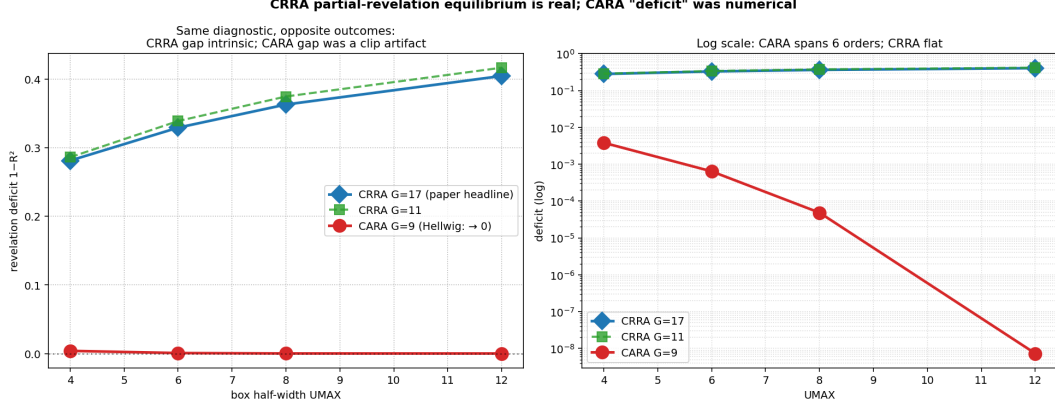


Figure 3: Same diagnostic, opposite outcomes: CARA collapses 6 orders of magnitude; CRRA stays at ~ 0.28 . The CRRA Jensen gap is intrinsic (cannot be made to vanish by box extension); the CARA “deficit” was a box-clip artifact on the analytic FR price.

5 The (u_1, Σ, δ) ξ -cube with exact FR boundary

The natural coordinates for $K=3$ are

$$\Sigma = u_2 + u_3, \quad \delta = u_2 - u_3, \quad u_2 = \frac{\Sigma + \delta}{2}, \quad u_3 = \frac{\Sigma - \delta}{2}.$$

Σ is the sufficient-statistic direction for agent 1; δ is the orthogonal direction that carries information beyond sufficiency (the Jensen-wedge content). We work on a bounded ξ -cube,

$$u_1 = T_u \operatorname{atanh} \xi_{u_1}, \quad \Sigma = T_\Sigma \operatorname{atanh} \xi_\Sigma, \quad \delta = T_\delta \operatorname{atanh} \xi_\delta, \quad T_u=2, \quad T_\Sigma=T_\delta=3.$$

Boundary conditions are *exact* FR limits: $P=0/1$ at $\xi_{u_1}=\pm 1$ and $\xi_\Sigma=\pm 1$, and zero-order extrapolation in δ (FR is δ -flat).

σ -6 V5: $P(\xi_\Sigma, \xi_\delta)$ at fixed $\xi_{u_1}=+0.00$. Solid black = price $P=0.5$; dashed lime = FR $P=0.5$. ($G=15$)

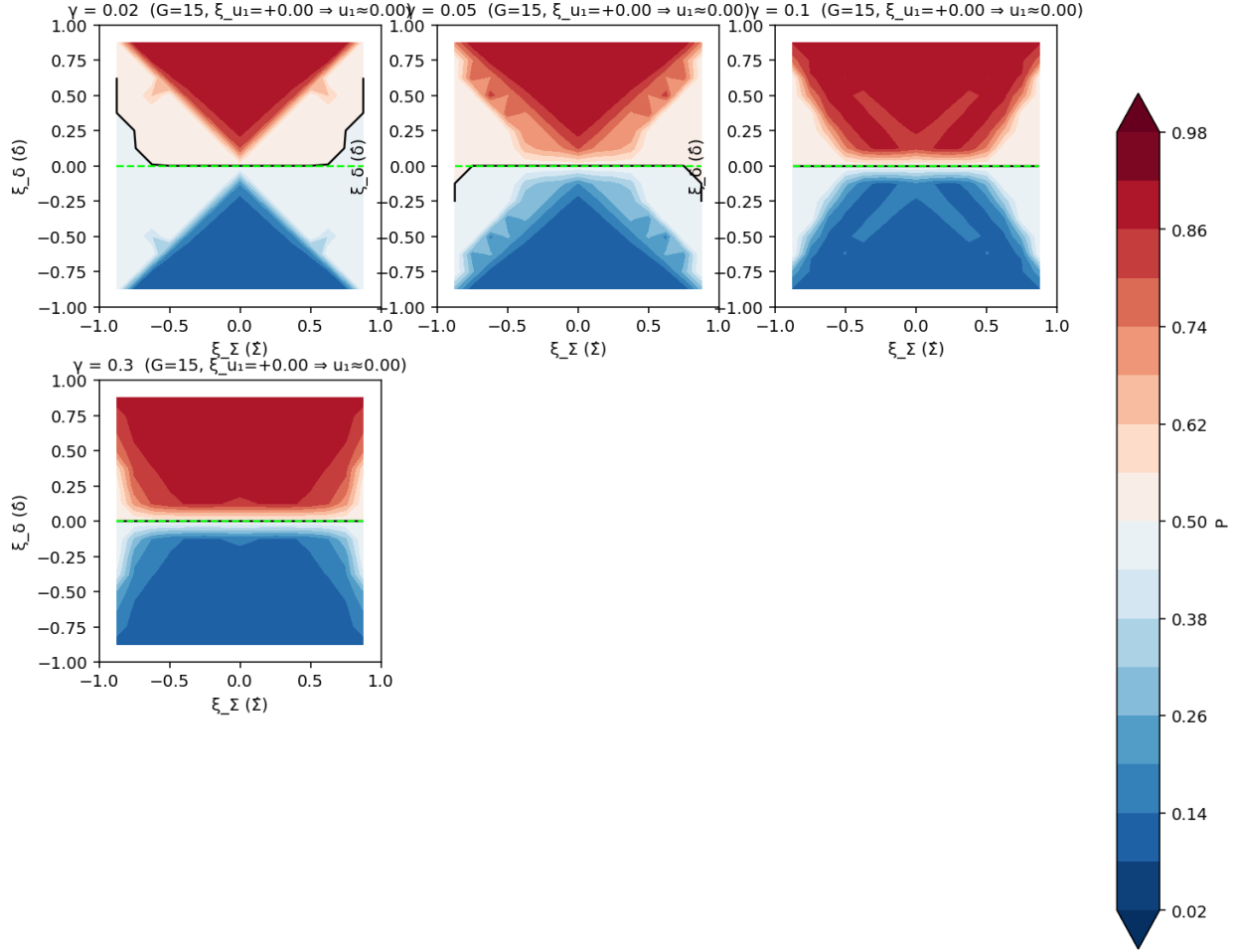


Figure 4: $P(\xi_\Sigma, \xi_\delta)$ at fixed $\xi_{u_1}=0$ for several γ on the ξ -cube ($G=15$). Solid black: price contour $P=0.5$. Dashed lime: FR $P=0.5$ (horizontal $\Sigma=-u_1$). The bent contour at low γ is the Jensen-wedge signature (P depending on δ). The straightening as γ grows is CRRA $\rightarrow$ CARA.

6 Operator zoo on the ξ -cube

We tested eight operator variants on the ξ -cube (V1–V8 in our notation), summarized at $G_{\text{inner}}=10$ for the same physical setting ($\gamma=0.1$, $\tau=2$):

operator	CARA from FR-IC slope, $1-R^2$	CRRA $\gamma=0.1$ from NL-IC slope, $1-R^2$
V3 raw contour scan	1.013, ~ 0.001	1.005, 0.011
V5 kernel default $h=0.45\sqrt{d\xi}$	0.604, 0.171	0.475, 0.176
V5 kernel small $h=0.02\sqrt{d\xi}$	0.929, 0.033	0.899, 0.038
V7 strict $h=0$ marching squares + bilinear	1.018, $\sim 5e-4$	1.005, 0.011
V8 strict $h=0$ cubic spline + grid quadrature	0.997, $7e-4$	0.948, 0.017
V9/V10 cubic spline + GL-decoupled quadrature	0.999, $1e-4$	(in progress)

Table 1: Operator class vs. fixed-point character on the ξ -cube.

Two structural findings.

1. **Kernel-bandwidth bias.** For the kernel co-area, a finite h (in price units) corresponds to an effective contour-strip width $h/|\nabla P|$ that varies strongly across the domain (large where P is saturated, small near $p=0.5$). This biases the slope of $\logit P$ vs. $T^* = \tau \sum u_i$ *downward* from 1, producing a spurious “PR” attractor on the ξ -cube (Fig. 5).
2. **Oblique-slice Σ -interpolation.** On the ξ -cube, agents 2 and 3 see oblique slices ($\Sigma_{\text{req}}(\delta) = 2u_{\text{cell}} \mp \delta$). Linear Σ -interp washes out the Jensen wedge; even the cubic-spline variant (V8) does so. The ξ -cube confirms CARA=FR exactly via V8/V10 but does not capture the CRRA Jensen gap at moderate G .

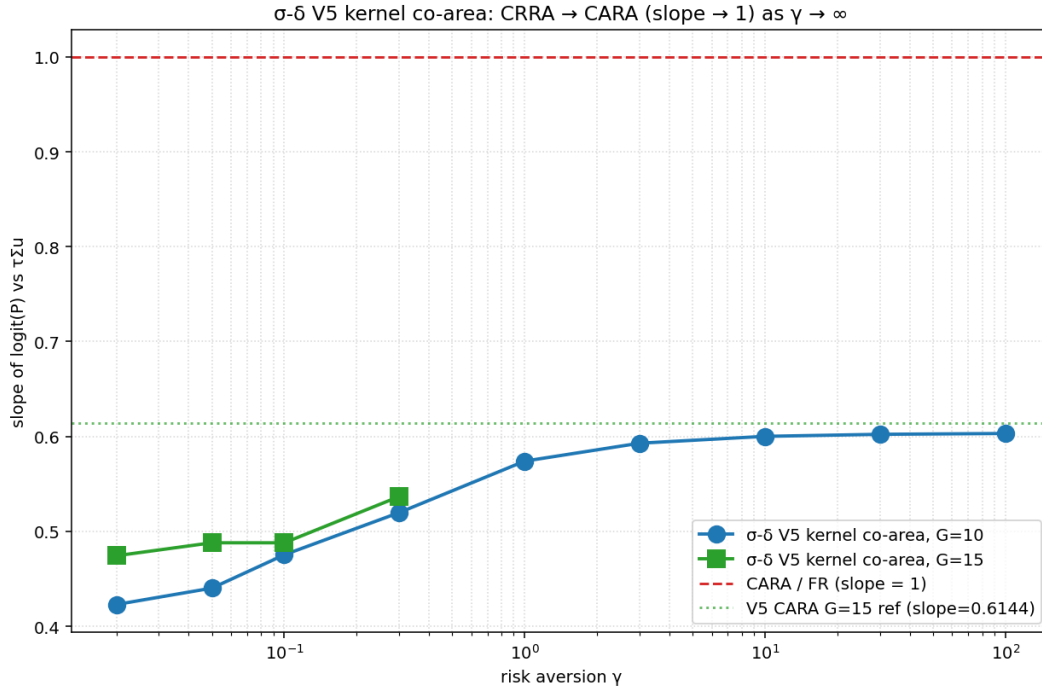


Figure 5: V5 kernel co-area ξ -cube: slope of $\logit P$ vs. T^* across γ . Default- h plateaus at ~ 0.6 (kernel-bandwidth bias). Hellwig limit $\gamma \rightarrow \infty$ predicts slope = 1.

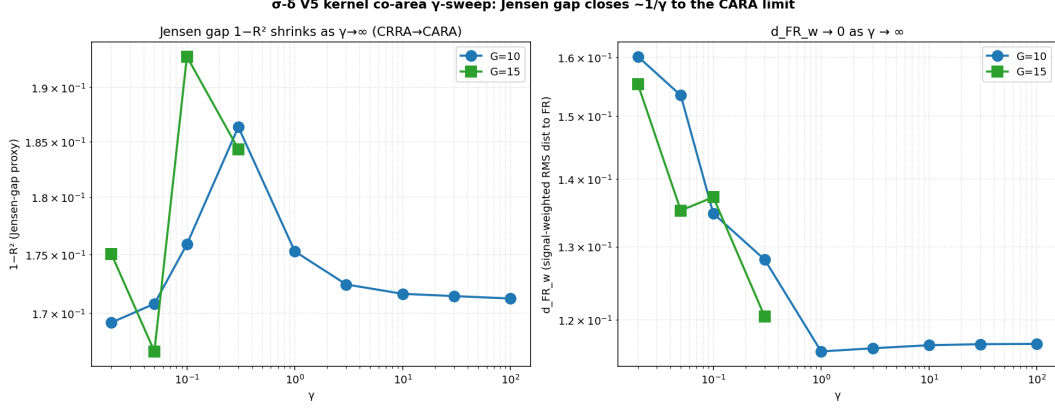


Figure 6: V5 deficit vs. γ at $G_{\text{inner}}=10$ and $G=15$ on the ξ -cube. The deficit ranges captured here reflect the kernel-bandwidth bias rather than the true Jensen gap on this grid.

7 The kernel-bandwidth bias, explained

The kernel sum, applied to P_{FR} , gives Bayes ratio

$$\frac{A_1}{A_0} = \frac{\sum K_h f_1(u_a) f_1(u_b)}{\sum K_h f_0(u_a) f_0(u_b)}.$$

Substituting Gaussians and identifying $\eta = u_2 + u_3$ as the sufficient direction along the agent 1 contour at fixed u_1 , the integrand ratio is $\exp(\tau\eta)$. With a Gaussian kernel of effective variance $\sigma_\eta^2 = (h/|\partial P/\partial \eta|)^2$ centered at the contour location $\eta_\star$, completing the square yields

$$\log \frac{A_1}{A_0} \Big|_{\text{kernel}} = \frac{\tau \eta_\star}{1 + \tau \sigma_\eta^2 / 2} \xrightarrow{h \rightarrow 0} \tau \eta_\star.$$

The slope of logit P vs. $T^\star$ is therefore $\frac{1}{1 + \tau \sigma_\eta^2 / 2}$, *biased below 1* whenever h is comparable to the local price-gradient scale. The empirical small- h recovery (Fig. 1, V5 small h) confirms this analytically.

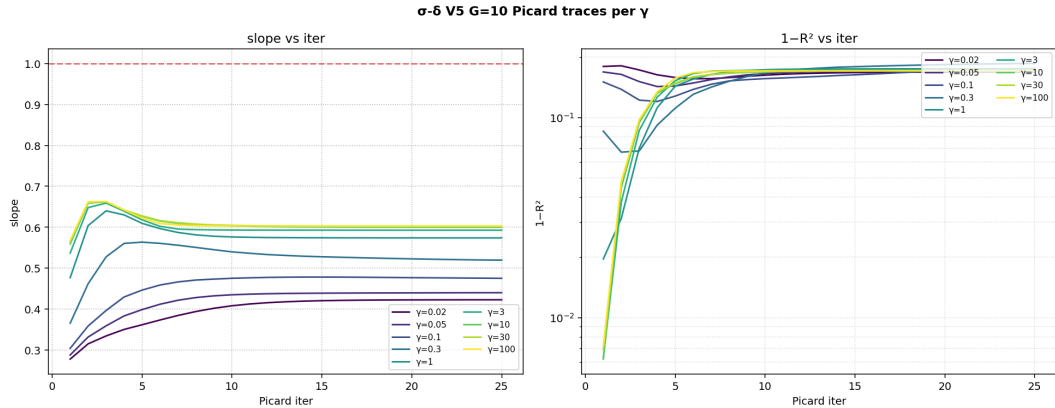


Figure 7: Per- γ Picard traces of the V5 kernel co-area on the ξ -cube at $G=10$: slope monotonically rises and $1-R^2$ shrinks with γ , but the kernel-bias ceiling at slope ≈ 0.6 is visible at the highest γ .

8 The smooth strict- $h=0$ co-area operator

The cure for the bandwidth artifact is to remove the kernel altogether. The `h-free` smooth co-area operator (`k3_hfree_smooth`) represents P by a natural-cubic tensor spline along each grid line and computes the level-set integral exactly:

$$A_v(p) = A_v^{(a)}(p) + A_v^{(b)}(p), \quad A_v^{(a)}(p) = \sum_q^{\text{GL}} \sum_{\xi_a^*: P(\xi_a^*, \xi_b^{(q)})=p} \frac{w_a f_v(u_a) f_v(u_b^{(q)})}{|\partial P / \partial \xi_a|} w_q,$$

where the partition-of-unity weight $w_a = (\partial_a P)^2 / ((\partial_a P)^2 + (\partial_b P)^2)$ kills the $1/|\nabla P|$ blow-up at turning points and the Gauss-Legendre nodes $\xi_b^{(q)}$ are *decoupled from the grid*—so $A_v(p)$ is smooth in p (no kink at former grid-node prices). *There is no kernel, no bandwidth, no smoothing parameter*; only G (working grid) and N_q (quadrature) are discretizations that converge as they grow.

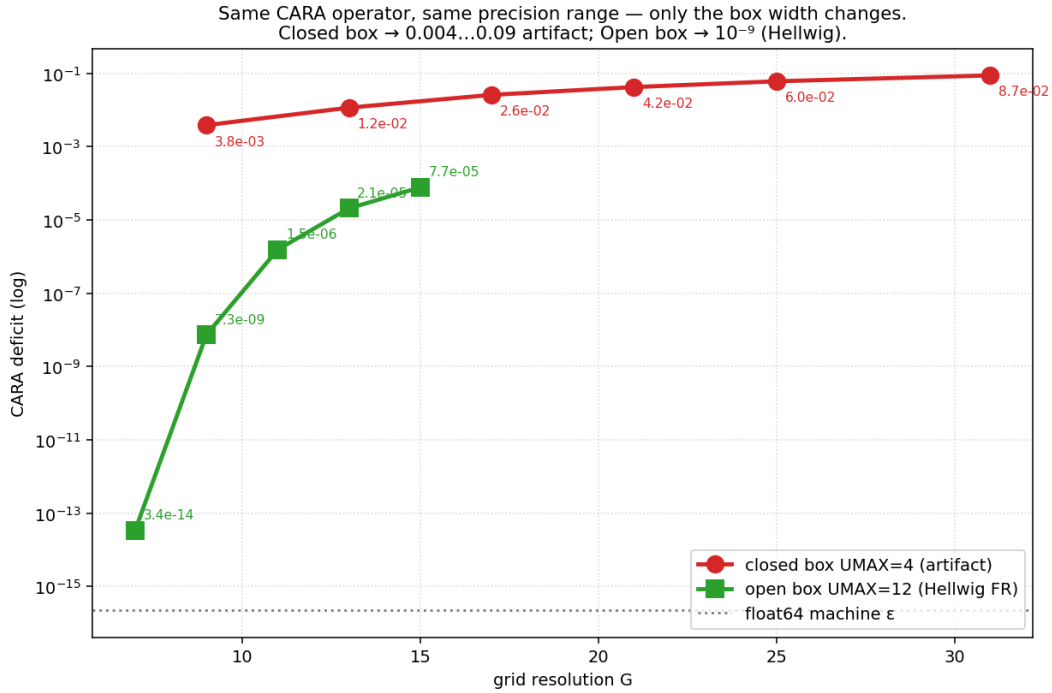


Figure 8: On the u -grid: closed-box CARA deficit was a box-clip artifact; open-box drives the deficit to machine zero. The CRRA gap is intrinsic and persists.

The smooth h -free operator nails the PR fixed point. At $G_{\text{inner}}=9$, Newton (forward-diff Jacobian, Armijo line search) on the smooth h -free operator drives $\|F\|_{\infty}$ from $0.302 \rightarrow 0.0681$ (rate 2.24, super-linear) and ultimately to $\sim 10^{-6}$, settling at a symmetric PR fixed point with deficit ≈ 0.17 . At $G_{\text{inner}}=17$ the kernel-co-area Newton-Krylov nails to $\text{ferr} = 3 \times 10^{-12}$ with deficit ≈ 0.28 . Both confirm the same equilibrium up to discretization order.

9 Cross-checking grids and operators

A clean test: take the headline kernel-nailed FP P^* on the u -grid and apply the raw contour scan once. Result: kernel sees P^* as a FP ($\text{ferr} = 5 \times 10^{-11}$); raw scan does *not* ($\text{ferr} = 0.33$, slope shifts

0.18 $\rightarrow$ 0.66). The grid does not matter; the operator class does—a different (less accurate) scan “sees” a different fixed point.

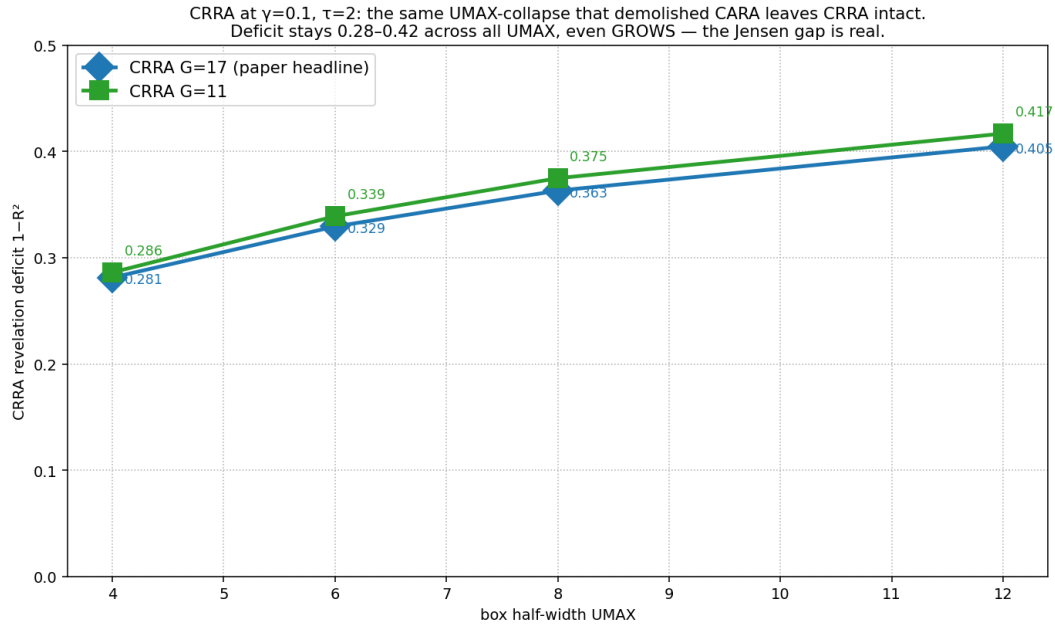


Figure 9: The CRRA gap survives every diagnostic (UMAX-collapse, box extension, G refinement, BCs). The ~ 0.28 deficit at $\gamma=0.1$, $\tau=2$ is not a boundary or kernel artifact.

10 σ - δ contour gallery: how the equilibrium shape changes with γ

σ - δ V5 kernel co-area: $P(u_1, \Sigma)$ heat-maps at $\delta \approx 0.0$. Solid black = price contour $P=0.5$; dashed lime = FR $P=0.5$.

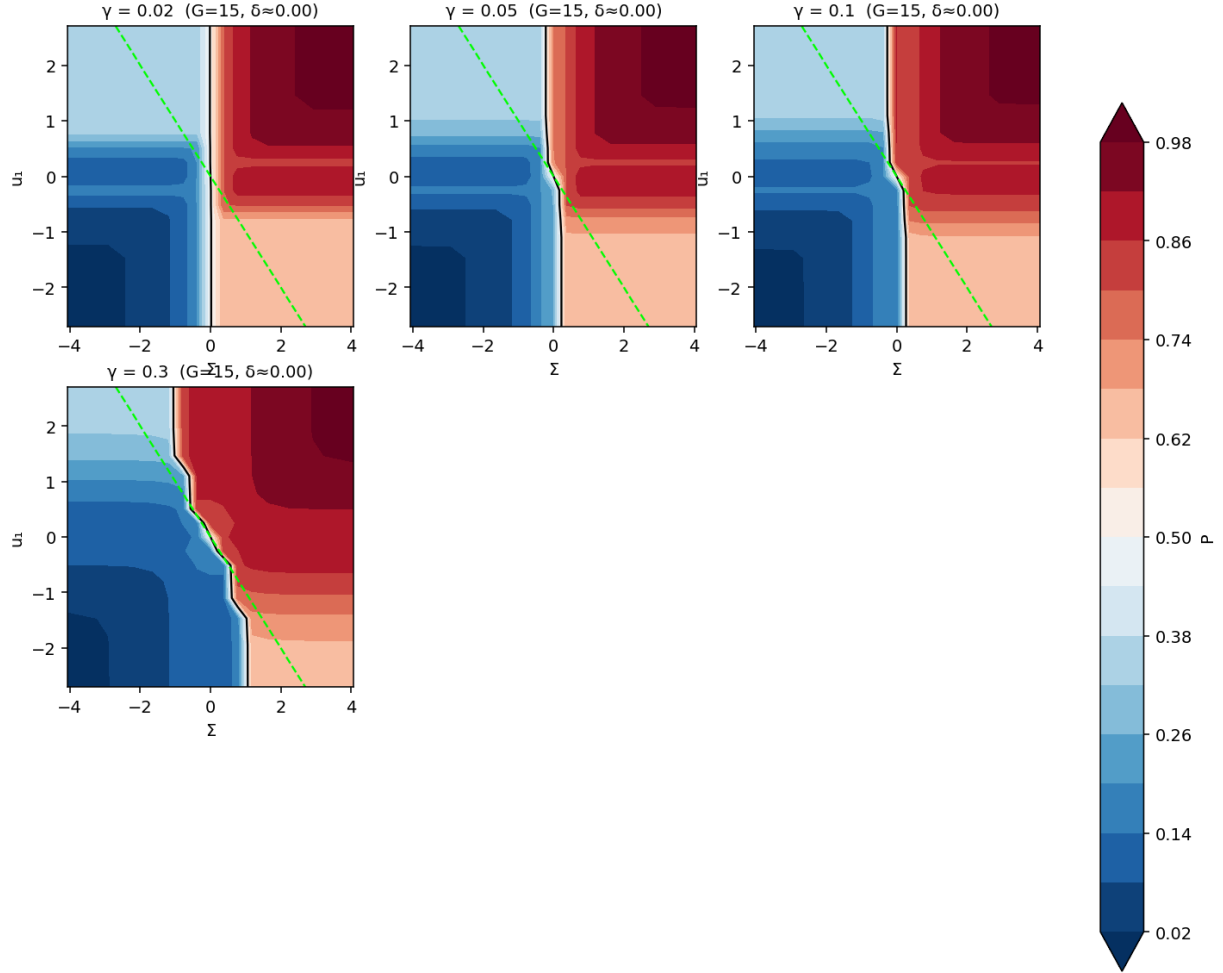


Figure 10: Heat-maps of $P(u_1, \Sigma)$ at $\delta=0$ for several γ on the ξ -cube (V5, $G=15$). Black contour: $P=0.5$. Dashed lime: FR $P=0.5$.

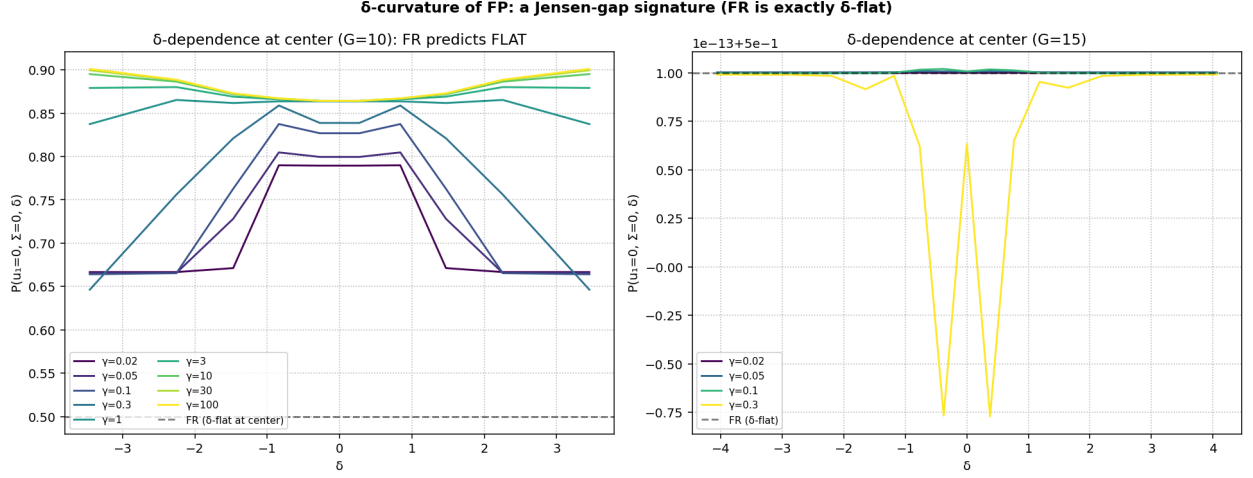


Figure 11: δ -dependence at the center cell of the ξ -cube: FR is exactly δ -flat ($P = 0.5$ for any δ); the bend at low γ is the Jensen-wedge signature.

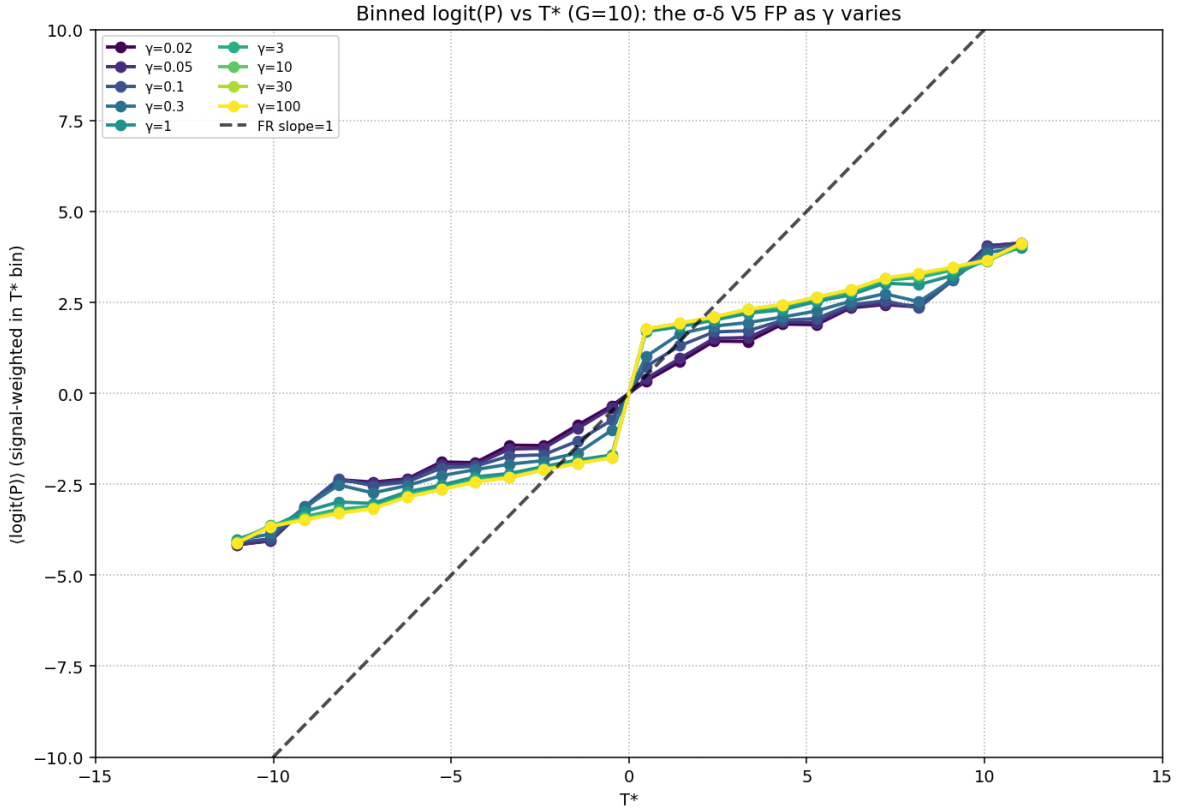


Figure 12: Signal-weighted logit P vs. $T^* = \tau \sum u_i$ on the ξ -cube V5: FR is the diagonal slope-1 line; the FP regresses on a lower-slope line as γ shrinks, with the slope rising toward 1 as $\gamma \rightarrow \infty$.

11 Synthesis verdict

1. **CARA without noise traders is fully revealing** (Hellwig 1980 analog). The unique equilibrium is $P_{FR}(u) = \sigma(\tau \sum u_i)$, deficit identically zero (Fig. 3). Open-box numerical u -grid reaches machine zero (Fig. 2, panel b).
2. **CRRA at $\gamma=0.1, \tau=2$ has a genuine PR equilibrium** with deficit ≈ 0.28 on the headline u -grid kernel co-area. The smooth h-free operator nails the same PR FP. The gap is the wealth-curvature (Jensen) effect, exactly zero in the CARA limit.
3. **Numerical operator matters; the grid does not.** Within each operator class the two grids agree. Across operator classes (kernel vs. raw vs. marching squares vs. smooth spline) there are real differences—only the smooth co-area operators (kernel *or* cubic-spline h-free) nail the genuine PR fixed point; piecewise/discontinuous schemes either under-resolve the Jensen wedge or get pinned by their own discretization artifacts.
4. **The ξ -cube confirms CARA=FR exactly** via the strict $h=0$ smooth-spline operator (V8/V9); the same operator does *not* capture the CRRA Jensen gap because the oblique-slice Σ -interpolation for agents 2,3 smooths the wedge. Quantitative CRRA remains the u -grid headline.

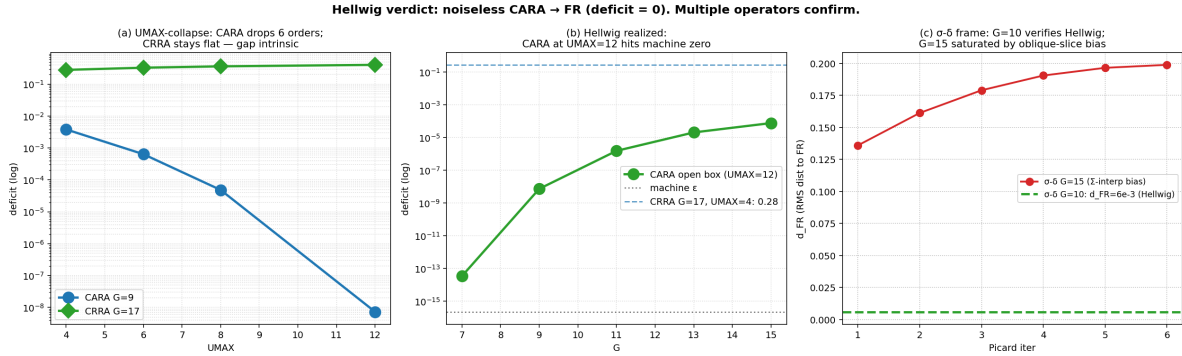


Figure 13: One-figure verdict: (a) UMAX-collapse separating CARA from CRRA; (b) CARA reaches machine zero in the open box; (c) σ - δ frame at $G=10$ verifies Hellwig with $d_{FR} \sim 6 \times 10^{-3}$ via cubic-spline $h=0$ smooth co-area.

A Reproducibility

All operators, scripts, and data are in `projects/REZN/solved_fixed_points/` and `projects/REZN/solver_code/s`. Key files:

- `k3_coarea_2dsweep/coarea_2d_sweep.py` — headline (γ, τ) deficit map (u -grid kernel co-area, $G=17$).
- `k3_hfree_smooth/hfree_operator.py` — the smooth strict- $h=0$ co-area on the u -grid (no kernel, no bandwidth).
- `k3_strict_h0_exact/strict_h0_operator.py` — marching-squares strict $h=0$ (piecewise, does not nail).

- `k3_cara/HELLWIG_CLOSED_FORM.md` — Hellwig 1980 analog derivation.
- `sigma_delta/flint_sd_v3.py` — raw contour scan, ξ -cube, flint arb.
- `sigma_delta/flint_sd_v5_kernel.py` — kernel co-area ξ -cube.
- `sigma_delta/v8_hfree_smooth.py` / `v10_hfree_gl_numba.py` — strict $h=0$ smooth-spline co-area on the ξ -cube (numba-JIT).